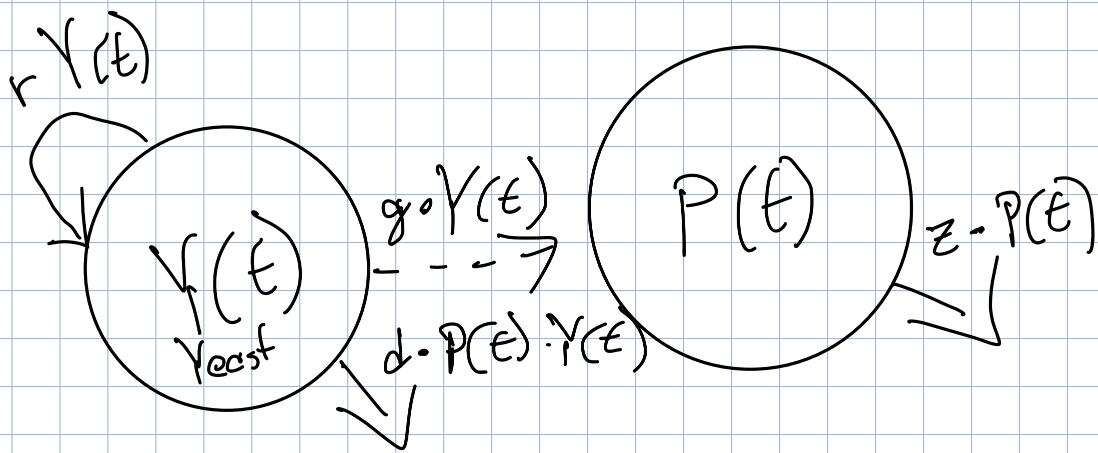
$0 < g \leq 1$ $\qquad$ $0 < r \leq 1$

$0 < d \leq 1$

$0 < z \leq 1$



$$\frac{dY}{dt} = rY - dPY \qquad \frac{dP}{dt} = gY - zP$$

$$g\hat{Y} - z\hat{P} = 0$$

$$\hat{P} = \frac{g}{z}\hat{Y}$$

$$0 = rY - dPY \rightarrow \boxed{\hat{Y} = 0} \rightarrow \boxed{\hat{P} = 0}$$

$$0 = r - dP$$

$$0 = r - \frac{dg}{z}Y$$

$$\boxed{\hat{Y} = \frac{rz}{dg}} \rightarrow \boxed{\hat{P} = \frac{r}{d}}$$

Jacobian Matrix

$$J = \begin{pmatrix} \frac{dY}{dY} & \frac{dY}{dP} \\ \frac{dP}{dY} & \frac{dP}{dP} \end{pmatrix} = \begin{pmatrix} r - dP & -dY \\ g & -z \end{pmatrix}$$

$$J\Big|_{\hat{P}=\frac{r}{d},\ \hat{Y}=\frac{rz}{d\gamma}} = \begin{pmatrix} 0 & -\frac{rz}{\gamma} \\ \gamma & -z \end{pmatrix}$$

$$\lambda_1 = \frac{-\gamma z - \sqrt{\gamma^2 z(z-4r)}}{2\gamma} \qquad \lambda_2 = \frac{-\gamma z + \sqrt{\gamma^2 z(z-4r)}}{2\gamma}$$

$$\gamma > 0 \Rightarrow \lambda_1 = \frac{-z - \sqrt{z(z-4r)}}{2} \qquad \lambda_2 = \frac{-z + \sqrt{z(z-4r)}}{2}$$

Stability independent on d or γ

Oscillations: possible if $\sqrt{z(z-4r)} < 0$

$$z^2 - 4rz < 0 \qquad | : z$$

$$z - 4r < 0$$

$$\boxed{z < 4r} \rightarrow \boxed{\frac{z}{r} < 4}$$

z has to be more than 4-fold of r

If oscillations, equilibrium is stable:

$$Re(\lambda_{1/2}) = -\frac{z}{2}$$

since $4r > 0$, the term on the left can only become smaller

if no oscillations: $z(z-4r) > 0$ $\rightarrow z > 4r$

$\rightarrow z - 4r > 0$

$z - 4r < z$

equilibria is stable when

$$\frac{-z + \sqrt{z(z-4r)}}{2} < 0 \quad (\lambda_2)$$

$-z + \sqrt{z(z-4r)} < 0$

$z > 0$ per definition $\rightarrow -z < 0$

$\sqrt{z(z-4r)} < z \quad | \, x^2$

$z(z-4r) < z^2$

$z^2 - 4zr < z^2 \quad | - z^2$

$-4zr < 0 \quad \forall z, r$ always true

$$\frac{-z - \sqrt{z(z-4r)}}{2} < 0 \quad (\lambda_1)$$

$-z - \sqrt{z(z-4r)} < 0$

$z > 0$ per definition $\rightarrow -z < 0$

$\sqrt{z(z-4r)} > 0 \; \forall z \geq 4r \rightarrow -\sqrt{z^2 - 4zr} < 0$

always negative!

$\rightarrow$ Conclusion: equilibrium always stable, oscillation-free for $z \geq 4r$